Assessing Agricultural Drought Using Remote Sensing and GIS: A Case Study of Afghanistan

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1 Introduction

The impacts of climate change and global warming have begun to become visible in all aspects of the environment, from the water cycle to the soil temperature. The direct implications of all of these are seen in agriculture and subsequently in food security. The decrease in precipitation and the erratic rainfall have caused the area of drought affected regions to increase at an alarming rate. The impact of severe and extreme drought on the agricultural zone is threatening to the entire population. Assessing agricultural drought is essential to understand its impact on food security, particularly in drought-prone regions such as Afghanistan, especially when a large group of the population depends on agriculture for their livelihood. To address this issue, the Food and Agriculture Organization (FAO) conducted a comprehensive analysis that incorporated Remote Sensing and Geographic Information Systems (GIS) to evaluate the severity of drought and its effects on agricultural systems.

2 Background and Objective

This longitudinal project was implemented by the FAO to assess the growing impact of agricultural drought in Afghanistan. These findings are important for the Afghanistan Emergency Food Security Project (OSRO/AFG /213/WBK), which provides essential information to address food security challenges. It also helps to implement appropriate and timely interventions to reduce the severity of the implications.

3 Data and Tools

Satellite-based Remote Sensing data were used to assess vegetation health and drought conditions, specifically the Vegetation Condition Index (VCI) from MODIS product. Areas with VCI values below 40 percent were classified as drought-affected. In addition, land cover datasets from the International Centre for Integrated Mountain Development (ICIMOD, 2018) were used to extract rainfed and irrigated agricultural areas.

4 Methodology

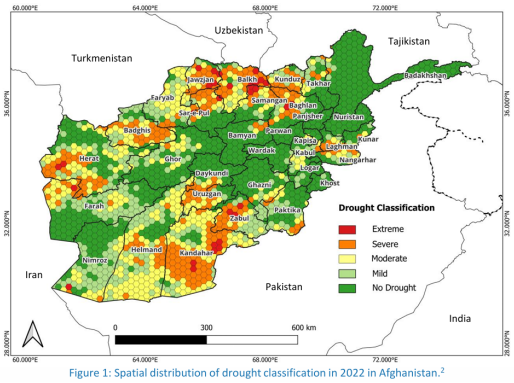
To assess drought severity, monthly VCI maps from January to April were merged to create a comprehensive map of the growing-season, which was then categorized into five drought severity classes. This map was then visualized using a hexagonal grid with cells of approximately 350 km^2.

To evaluate the exposure of rainfed and irrigated agriculture to drought, GIS techniques were used where drought data was overlayed with rainfed and irrigated agriculture maps to evaluate the level of exposure in different provinces. The results were visualized using hexagonal grid mapping (90 km^2 per hexagon), where each grid cell represented a defined spatial area and was classified into different exposure levels ranging from low to high.

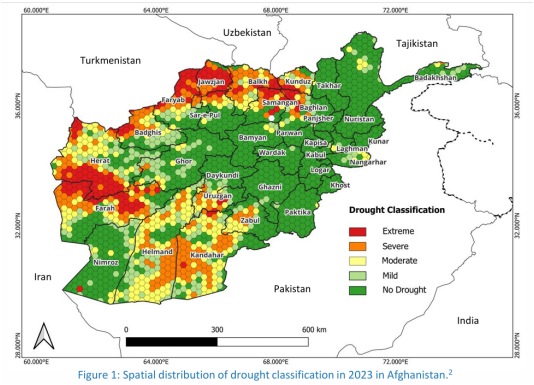
5 Key Outcomes

In recent years, there has been colossal degradation of agricultural land. This project tried to demonstrate the severity of the degradation by comparing the spatial analysis of successive years.

The main outcome of this project was to provide visual evidence of agricultural drought in Afghanistan and provide essential information to the Afghanistan Emergency Food Security Project (OSRO/AFG/213/WBK) so that food security challenges are addressed. It also urged the associated stakeholders to carry out proper and timely interventions.



(a) FAO, 2023[1]



(b) FAO, 2023[2]

5.1 Rainfed Agriculture Results

The findings showed that a significant proportion of rainfed agriculture in Afghanistan were affected by severe drought conditions in 2023, with a serious impact observed in the western and northern regions. Herat Province experienced the largest affected area by total drought (2,207 km^2), while Samangan Province was hit hardest by extreme drought (492 km^2). Compared to 2022, Samangan saw an increase of 291 km^2 in extreme drought, demonstrating the severity of the situation.

5.2 Irrigated Agriculture Results

In comparison, irrigated agriculture also experienced considerable drought impacts, especially in the northern, western, and southern regions. Although irrigation systems provided some level of resilience, these areas were still affected by severe drought conditions. In 2023, Balkh Province had the highest total drought area where 1,259 km^2 of the area was affected by total drought which was 191 km^2 increase from 2022. On the other hand, Jawzjan had 419 km^2 of area affected by extreme drought in 2023, which is an increase of 171 km^2 compared to 2022.

5.3 Drought Severity Analysis

The drought severity analysis was conducted by combining monthly VCI data from January to April to represent the growing season. The results were categorized into five classes: extreme, severe, moderate, mild, and no drought. The 2023 analysis indicated that 11% of the areas experienced extreme drought, 15% severe drought, 14% moderate drought and 13% mild drought, while 47% of the areas did not show drought conditions. The northern and western regions were consistently identified as the most affected. Herat Province saw an increase of 8% in extreme drought in one year. This highlights that the situation is in dire need of immediate intervention.

6 Challenges/ Limitations

Since the analysis was performed on the basis of the satellite datasets, it lacks ground validation data to confirm the satellite results. The assessment is largely dependent on the availability and quality of the datasets, which might affect the analysis comprising the other variables. There is a lack of comparison to a larger timeframe due to the recent implementation of the project.

7 Application in Nepal

Nepal, too, is facing the increasing implications of climate change and global warming. Nepal is situated in a vulnerable geographical region, where it is in risk of both flood and drought, which directly impacts agriculture and subsequently food security.

Nepal is in dire need of constant monitoring to limit the consequences of climate related disasters. According to the National Sample Census of Agriculture 2021/22 made public by the National Statistics Office, about 4.13 million households are involved in agriculture, which constitutes 62% of the total households and 67% of the total population, whose livelihood can be directly impacted by any of the climate related hazards.

Hamal *et al.* (2021)[7] stated that the cumulative frequency of short-term droughts for the Western, Central and Eastern regions of Nepal increased by 15%, 18%, and 15%, respectively, from 2005 to 2017. This raises an important question regarding how the drought is being monitored and what interventions have been implemented to overcome the said situation. To address these challenges, remote sensing and GIS can be implemented to routinely monitor and provide timely interventions, thereby minimizing the risk of severe impact.

References

[1] Food and Agriculture Organization. Assessment of agricultural drought severity in afghanistan for 2022. Technical Report CC8265EN/1/10.23, FAO, Rome, Italy, 2023.

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[4] Food and Agriculture Organization. Assessing exposure of irrigated agriculture to drought in afghanistan (2023). Technical Report CC9766EN/1/02.24, FAO, Rome, Italy, 2024.

[5] Food and Agriculture Organization. Assessing exposure of rainfed agriculture to drought in afghanistan (2022). Technical Report CC9493EN/1/01.24, FAO, Rome, Italy, 2024.

[6] Food and Agriculture Organization. Assessing exposure of rainfed agriculture to drought in afghanistan (2023). Technical Report CC9766EN/1/02.24, FAO, Rome, Italy, 2024.

[7] Kalpana Hamal, Shankar Sharma, Binod Pokharel, Dibas Shrestha, Rocky Talchabhadel, Alen Shrestha, and Nitesh Khadka. Changing pattern of drought in nepal and associated atmospheric circulation. *Atmospheric Research*, 262:105798, 2021.

[1] [2] [3] [4] [5] [6] [7]